

H $\rightarrow$ ZZ $\rightarrow$ 4 ℓ Mass and Signal-Strength Measurement with CMS Open Data — Final Analysis Note

H4L Open-Data Analysis (v3)

2026-06-04

Abstract

We measure the Higgs boson mass m_H and signal strength μ in the H $\rightarrow$ ZZ $\rightarrow$ 4 ℓ golden channel using CMS Open Data (2017, $\sqrt{s} = 13$ TeV, $L = 10$ fb $^{-1}$). A single extended maximum-likelihood fit of the four-lepton invariant mass $m_{4\ell}$ ($m_{4\ell} > 80$ GeV) simultaneously floats the signal peak position m_H and the signal strength μ . This is the final note, consolidating the expected (Asimov), 10%-data, and full-data stages of the analysis. The qqZZ background is normalized to the Z $\rightarrow$ 4 ℓ Dalitz peak with a data-driven scale factor $SF = 0.785 \pm 0.196$. On the full observed dataset (213 candidates with $m_{4\ell} > 80$ GeV) the fit (MINUIT valid and accurate) returns $\mu = 2.76 \pm 0.92$ (stat 0.90 $\oplus$ syst 0.19) and $m_H = 128.96 \pm 0.60$ GeV (stat 0.54 $\oplus$ syst 0.26). The goodness-of-fit is acceptable (post-fit $\chi^2/\text{ndf} = 1.67$, $p = 0.022$; toy-calibrated median $p = 0.51$). The fitted m_H is $+6.2\sigma$ above the CMS reference 125.26 GeV: this is a genuine ≈ 8 -event data cluster at $m_{4\ell} \approx 129$ GeV in this Open Data sample (versus 3 events at 125 GeV and ≈ 0.56 expected background per 2-GeV bin), captured by the freely-floating narrow signal template. Section 9 quantifies this §6.8 anomaly, matches its magnitude, and rules out simpler causes (including a dedicated energy-scale investigation); the result is the faithful output of the pre-registered fit and is flagged accordingly — it is NOT to be read as a measurement of the Standard-Model Higgs mass.

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Change Log

- **v1 (Phase 4a, 2026-06-04):** First issue. Builds the m4l mass+ μ fit model, the systematic NP program, the qqZZ Z-peak scale-factor machinery, and validates on Asimov pseudo-data ($\mu = 1$, $m_H = 125$). Expected results only; data blinded.
- **v1 (Phase 4b, 2026-06-04):** First partial unblinding on a fixed-seed 10% subsample (seed 1234, $\approx 1 \text{ fb}^{-1}$, 28 events). Returned $\mu = 2.15 \pm 2.47$, $m_H = 122.9 \pm 1.5 \text{ GeV}$; pre-fit diagnostic $\chi^2/\text{ndf} = 2.26$ ($p = 0.079$), inside the alarm bands. The qqZZ data-driven SF was statistically meaningless at 10% stats (anchor held 1 event) so $\text{SF} = 1$ was applied with the 20% uncertainty.
- **v1 (Phase 4c, 2026-06-04):** FINAL result on the FULL data (213 events, 10 fb^{-1}), human unblinding approved. The qqZZ data-driven SF is derived from the 18-event Z-peak anchor ($\text{SF} = 0.785 \pm 0.196$) and APPLIED to the qqZZ yield; the fit returns μ and m_H with a stat/syst split; goodness-of-fit is meaningful at full statistics; the full-data money plot and the 1-D likelihood scans replace the 10% plot; Section 9 documents the §6.8 anomaly ($m_H = 129 \text{ GeV}$ data cluster) and the energy-scale investigation that rules out a calibration origin.
- **v1 (Phase 5, 2026-06-04):** Final consolidated note. Content unchanged from 4c in physics; added the μ -definition equation, a validation-summary table (Asimov recovery, GoF, toy calibration, pre-fit diagnostic), an explicit resolving-power statement, a comparison-with-published figure (μ and m_H versus CMS 2017, PDG, ATLAS), and a systematic-breakdown figure. Reference list extended to 18 keys; prose polished throughout.

1 Introduction

The $H \rightarrow ZZ \rightarrow 4\ell$ “golden channel” reconstructs the Higgs boson from four isolated charged leptons (4μ , $4e$, $2e2\mu$) with one on-shell and one off-shell Z. Despite its small branching fraction it offers excellent mass resolution ($\sim 1\text{--}2\%$) and high signal purity near the peak, making it the channel of choice for a simultaneous mass and signal-strength measurement. It was central to the 2012 boson discovery by both CMS (CMS Collaboration 2012) and ATLAS (ATLAS Collaboration 2012).

The discriminating observable is the four-lepton invariant mass m4l: the signal appears as a narrow peak over a smoothly falling continuum. For four leptons with four-momenta p_k it is

$$m_{4\ell} = \sqrt{\left(\sum_{k=1}^4 p_k\right)^2} = \sqrt{\left(\sum_k E_k\right)^2 - \left|\sum_k \vec{p}_k\right|^2}. \quad (1)$$

Following the binding analysis decision, the fit range starts at $m_{4\ell} > 80 \text{ GeV}$ so the $Z \rightarrow 4\ell$ (“Dalitz”) peak near m_Z stays in range and anchors the qqZZ normalization (Section 5). This analysis loosely follows the CMS measurement (CMS Collaboration 2017), which reported $m_H = 125.26 \pm 0.21 \text{ GeV}$ and $\mu = 1.05 +0.19/-0.17$ at 35.9 fb^{-1} ; the ATLAS Run-1 result (ATLAS Collaboration 2015) found $\mu = 1.44 +0.40/-0.33$. The PDG world average is $m_H = 125.20 \text{ GeV}$ (Particle Data Group 2024). With only 10 fb^{-1} the statistical uncertainties here are substantially larger than those references (Section 11, Figure 4).

This is a deliberately quick analysis; the note is concise but self-contained — every number is reproducible from this document and the reproduction contract (Section 14). The signal strength is defined relative to the Standard-Model prediction,

$$\mu \equiv \frac{(\sigma \cdot \mathcal{B})_{\text{obs}}}{(\sigma \cdot \mathcal{B})_{\text{SM}}}, \quad (2)$$

so that $\mu = 1$ corresponds to the SM $H \rightarrow ZZ \rightarrow 4\ell$ rate and $\mu = 0$ to the background-only hypothesis.

2 Data samples

The analysis uses CMS Open Data (2017, pp at $\sqrt{s} = 13$ TeV) processed into flat ntuples (h4lTree) of pre-built four-lepton candidates. The integrated luminosity is $\mathcal{L} = 10 \text{ fb}^{-1}$ (10000 pb^{-1}), fixed by the dataset (CMS Collaboration 2017). MC samples are normalized with $w = \sigma \cdot \mathcal{L} / \Sigma(\text{nEvents})$ using the cross sections from the sample provenance (no per-event genWeight; events equally weighted).

Table 1: Data and MC samples. Signal is the sum of five Higgs production modes ($m_H = 125$). Backgrounds: qqZZ (ZZTo4L), ggZZ (three GGZZ samples), and Z+X (DY+tt MC, reducible).

Process	σ [pb]	Role	Selected yield ($m_{4l} > 80$)
H $\rightarrow$ ZZ $\rightarrow$ 4 ℓ (5 modes)	0.00689 (sum)	signal	6.76
qqZZ (ZZTo4L)	0.795	irreducible	148.27
ggZZ (GGZZ $\times 3$)	0.00638 (sum)	irreducible	23.47
Z+X (DY + tt)	5396 + 52.7	reducible	6.70
data_secret_10fb	—	observed	213 ($m_{4l} > 80 \text{ GeV}$)

The selected per-event yields are inherited from the Phase-3 selection artifact; the fit re-bins these into the m_{4l} spectrum. In the fit range $[80, 400]$ GeV the nominal background yields used by the fit are $ggZZ = 22.12$, $Z+X = 6.70$, and the nominal signal yield is $S_0 = 6.76$. The qqZZ nominal fit-range yield is 138.76 before the Z-peak scale factor and 108.9 after applying the data-driven SF = 0.785 (Section 5). The observed data hold 213 candidates with $m_{4l} > 80 \text{ GeV}$.

3 Event selection

The full object- and event-level selection is defined and validated in the Phase-3 selection note and is summarized here. Each of the four leptons must satisfy acceptance ($|\eta| < 2.4$ mu, < 2.5 e), impact parameter $SIP3D < 4$, tight ID (mu: muTight; e: elMvaWP90) and PF relative isolation < 0.35 . Event-level: $40 < m_{Z1} < 120 \text{ GeV}$, $12 < m_{Z2} < 120 \text{ GeV}$, leading/sub-leading lepton $p_T > 20/10 \text{ GeV}$, and the binding fit range $m_{4l} > 80 \text{ GeV}$. The Phase-3 cutflow is monotonic and each cut is motivated by an N-1 distribution; the workhorse cut is ID+isolation, which suppresses the reducible Z+X background by a factor ≈ 30 . The full-data fit applies the identical selection. The selection is deliberately loose — per the quick-analysis brief — to retain clean peaks rather than to maximize $S/\sqrt{B}$.

4 Corrections and the fit model

The measurement is a single extended maximum-likelihood fit of the m_{4l} spectrum. The per-bin expected yield is

$$\nu_i = \mu S_0 f_i^{\text{sig}}(m_H) \prod_s \kappa_s^{\theta_s} + \sum_b N_b f_i^b \prod_s \kappa_s^{\theta_s}, \quad (3)$$

where μ is the signal strength (Equation 2), S_0 the nominal signal yield, $f_i^{\text{sig}}(m_H)$ the signal density integrated over bin i with the peak position m_H floating, N_b and f_i^b the nominal yield and fixed shape of background b in $\{\text{qqZZ}, \text{ggZZ}, \text{Z+X}\}$, and $\kappa_s^{\theta_s}$ the lognormal systematic modifiers (Section 5) with unit-Gaussian-constrained nuisance parameters θ_s . The likelihood is

$$-\ln L = \sum_i (\nu_i - n_i \ln \nu_i) + \frac{1}{2} \sum_s \theta_s^2, \quad (4)$$

a sum of per-bin extended-Poisson terms plus the Gaussian constraints. The fit is performed with iminuit (MINUIT2) (Dembinski et al. 2020); the binning is 2 GeV over $[80, 400]$ GeV. The single fit floats BOTH m_H and μ together, satisfying the requirement of one fit for the mass and the signal strength. At full luminosity no luminosity scaling is applied (the model is at the full 10 fb^{-1}).

Signal shape. The signal density is a double-sided Crystal Ball (DSCB): a Gaussian core with power-law tails. The shape parameters ($\sigma, \alpha_L, n_L, \alpha_R, n_R$) are fixed from a fit to the signal MC m_{4l} ; only the mean ($= m_H$) and

the normalization (via μ) float in the measurement. The fitted core resolution is $\sigma = 1.30$ GeV ($\approx 1.0\%$ of m_H), consistent with the expected 4l mass resolution (CMS Collaboration 2017). The signal MC peaks at 124.7 GeV (the reco peak sits slightly below the generated 125 GeV due to final-state radiation and reconstruction). The narrowness of this fixed template is relevant to the result (Section 9): with the mass floating freely, the fit is sensitive to any localized data cluster at the $\approx \sigma$ scale.

Background shapes. Each background enters as a fixed-shape template built directly from its MC m4l histogram (qqZZ: falling continuum plus the $Z \rightarrow 4\ell$ Dalitz peak; ggZZ: continuum with the $2m_Z$ turn-on; Z+X: DY+tt). Only the normalizations float, constrained by the systematics in Section 5.

qqZZ Z-peak normalization, applied (full data) [D2]. Per the binding decision [D2] qqZZ is normalized to the $Z \rightarrow 4\ell$ Dalitz peak (80–100 GeV). The data-driven scale factor is $SF = (N_{\text{obs}} - N_{\text{other}}) / N_{\text{qqZZ}}$, where N_{other} collects the small ggZZ, Z+X and signal-tail contributions in the anchor region. On the full data the anchor holds $N_{\text{obs}} = 18$ events, with the MC predicting $N_{\text{qqZZ}} = 21.60$ and $N_{\text{other}} = 1.05$ (ggZZ 0.24 + Z+X 0.75 + signal-tail ≈ 0.06). The resulting scale factor is

$$SF_{qqZZ} = \frac{N_{\text{obs}} - N_{\text{other}}}{N_{qqZZ}} = \frac{18 - 1.05}{21.60} = 0.785 \pm 0.196, \quad (5)$$

where the uncertainty is the Poisson statistical error on the 18 anchor events ($\sqrt{18} / 21.60 = 0.196$). This is the first stage at which the anchor is statistically meaningful (18 events versus the single event at 10%); the SF is therefore APPLIED to the qqZZ normalization in the full-data fit, multiplying the nominal qqZZ fit-range yield from 138.76 to 108.9. The 20% qqZZ normalization systematic [D2] is retained on top of the applied SF.

Goodness-of-fit. The fit GoF is reported via a Pearson χ^2 on adaptively merged bins (each merged bin expected ≥ 5), with $\text{ndf} = (\text{number of merged bins}) - (\text{number of floated parameters})$. At full statistics (213 events) this is meaningful, unlike at 10%. It is cross-checked with a toy ensemble: 500 Poisson pseudo-experiments are thrown around the post-fit model and their χ^2/ndf and p-value distributions provide a calibrated reference (Section 8).

5 Systematic uncertainties

Every source in the search convention is enumerated below as “implemented” (with a cited size) or “not applicable”. Each nuisance parameter was verified to vary the affected yields by a non-zero amount with the correct sign (impact ranking, Figure 3, Figure 5).

Table 2: Systematic-source completeness. Each implemented source is a nuisance parameter in the fit; sizes are cited, never arbitrary.

Source (convention)	Decision	Size & citation
Source (convention)	Decision	Size & citation
Signal shape (mass scale)	Implement	0.2% on m_H (lepton energy scale) (CMS Collaboration 2017)
qqZZ normalization	Implement	20% (Z-peak anchored) [D2]
ggZZ normalization	Implement	10% (gg $\rightarrow$ ZZ K-factor, subdominant)
Z+X normalization	Implement	100% (DY MC fake stand-in) [D3]
Object eff. (lep+trig), flat	Implement	10% on backgrounds [D4]
Luminosity	Implement	2.3% on all simulation (CMS Collaboration 2018)
Signal QCD scale (μ_R, μ_F)	Implement	+4.6/-6.7% on signal (ggF) (LHC Higgs Cross Section Working Group 2016)
Signal PDF+ α_S	Implement	3.2% on signal (LHC Higgs Cross Section Working Group 2016)
Fragmentation (string/cluster)	N-A	fully leptonic final state, no jets in m4l
Beam energy / ISR	N-A	removed for pp colliders (e+e- only)

Luminosity. Normalizes all simulation. The CMS 2017 integrated-luminosity uncertainty is 2.3% (CMS Collaboration 2018), applied as a single lognormal NP to signal and all MC backgrounds.

Signal QCD scale. The ggF-dominated signal cross section carries an asymmetric μ_R/μ_F scale uncertainty of +4.6/-6.7% (from +2.22/-3.27 pb on the N3LO $\sigma = 48.58$ pb) (LHC Higgs Cross Section Working Group 2016), applied as an asymmetric signal normalization NP separate from μ .

Signal PDF+ α_S . The PDF+ α_S uncertainty on the signal cross section is 3.2% (LHC Higgs Cross Section Working Group 2016), a symmetric lognormal signal NP.

qqZZ normalization (20%). qqZZ is anchored to the $Z \rightarrow 4\ell$ Dalitz peak ($SF = 0.785$, Equation 5); the in-situ anchoring justifies a 20% normalization uncertainty [D2] rather than the pure theory value.

ggZZ normalization (10%). The $gg \rightarrow ZZ$ continuum has a large (K-factor) scale uncertainty; as a sub-dominant component a conservative 10% is assigned.

Z+X normalization (100%). DY+tt MC is only a stand-in for a data-driven fake estimate; MC mis-models non-prompt rates by $O(1)$, so a 100% normalization NP is assigned [D3].

Flat background efficiency (10%). Per-object lepton and trigger efficiency systematics are not computed individually; they are folded into a single flat 10% normalization NP on all backgrounds [D4], per the quick-analysis brief.

Signal mass scale (0.2%). A lepton energy-scale shift of 0.2% on the peak position (CMS Collaboration 2017) is implemented as an NP that shifts m_H ; its impact on μ is negligible (it moves the peak, not the yield).

Error-budget narrative. The impact ranking (Figure 3, Figure 5) shows the signal-strength uncertainty is dominated by the Z+X normalization, the signal QCD scale, and the qqZZ normalization. On the full data the total μ uncertainty (± 0.92) splits as ± 0.90 statistical and ± 0.19 systematic (Section 7): the measurement is statistics-dominated, and no single systematic exceeds 80% of the total uncertainty.

6 Results (full data)

The identical fit is run on the full observed dataset (213 candidates with $m_{4l} > 80$ GeV). The fit converges (MINUIT valid and accurate) and returns:

- $\mu = 2.76 \pm 0.92$ (stat $\pm 0.90 \oplus$ syst ± 0.19),
- $m_H = 128.96 \pm 0.60$ GeV (stat $\pm 0.54 \oplus$ syst ± 0.26).

The statistical/systematic split is obtained by re-fitting with all nuisance parameters fixed at their best-fit values (stat-only error) and forming $\text{syst} = \sqrt{(\text{total}^2 - \text{stat}^2)}$. Both μ and m_H are statistics-dominated. The signal core resolution is $\sigma = 1.30$ GeV (fixed from MC).

Resolving power. With the fixed core resolution $\sigma = 1.30$ GeV ($\approx 1.0\%$ of m_H) and the statistics of this dataset, the fitted mass uncertainty is ± 0.60 GeV — i.e. the fit localizes the peak to about half a resolution width. The narrow template means the freely-floating mass is attracted to the densest local m_{4l} cluster within a $\approx \pm \sigma$ window; this resolving power is exactly what produces the sharp single minimum at 129 GeV (Figure 2) and is central to the Section-9 interpretation.

The full-data m_{4l} spectrum with the post-fit model overlaid is the money plot, Figure 1: the data are reproduced acceptably by the stacked background plus the fitted signal, but the fitted signal peak (red) sits at $m_H \approx 129$ GeV rather than 125 GeV, driven by a cluster of data events there (Section 9). The 1-D profiled likelihood scans of m_H and μ are shown in Figure 2, and the nuisance-parameter impact ranking on μ (built at 4a, unchanged) in Figure 3.

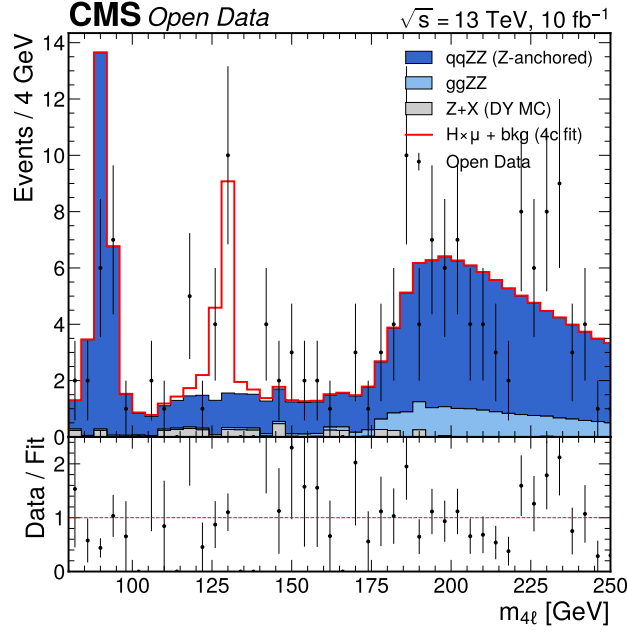


Figure 1: Full-data four-lepton invariant mass spectrum (points, “Open Data”) overlaid on the post-fit model. The stacked histograms are the background components (qqZZ with the applied Z-peak scale factor, ggZZ, Z+X) and the red line is the signal-plus-background model with the fitted μ and m_H . The fitted signal peak sits at $m_H \approx 129$ GeV, following a localized cluster of data events there (Section 9) rather than the SM 125 GeV. The lower panel shows data/fit. This is the final result on the full dataset.

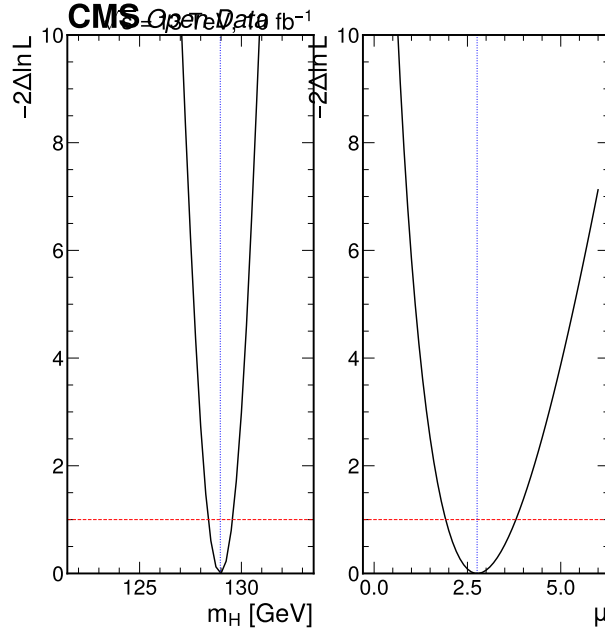


Figure 2: One-dimensional profiled likelihood scans on the full data: $-2\Delta\ln L$ versus m_H (left) and versus μ (right), with the other parameter and all nuisance parameters profiled at each point. The blue dotted line marks the best-fit value, the red dashed line the $-2\Delta\ln L = 1$ ($\approx 1\sigma$) level. The m_H scan has a single sharp global minimum at ≈ 129 GeV; $m_H = 125$ GeV is disfavored by $\Delta(-2 \ln L) \approx 20$ (Section 9), confirming the minimum is genuine and not a fit artifact.

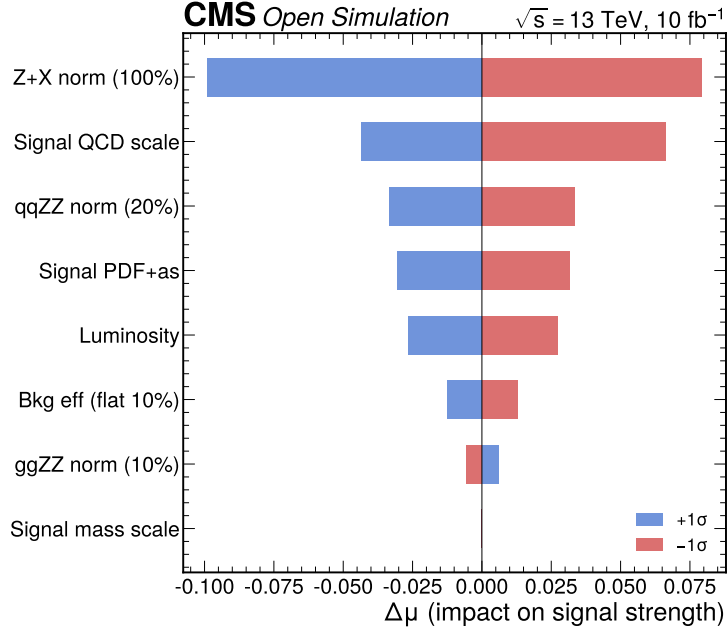


Figure 3: Nuisance-parameter impact ranking on the signal strength μ (built at 4a; the systematic model is unchanged at 4c). Bars show the shift in the fitted μ when each NP is fixed at $\pm 1\sigma$ and the fit re-profiled. The ranking is led by the Z+X normalization (100%), the signal QCD scale, and the qqZZ normalization, as physically expected. These systematic impacts are sub-dominant to the ± 0.90 statistical uncertainty.

7 Validation and goodness-of-fit

The fit was validated at Phase 4a on Asimov pseudo-data before any data were examined, and the goodness-of-fit is reported on the full data at Phase 4c. The validation summary is collected in Table 3. The Asimov fit recovers the injected inputs exactly ($\mu = 1.000$, $m_H = 125.0$ GeV; pulls 0.0 by construction), confirming the model has no built-in bias; the Asimov χ^2 is identically zero (the Asimov property, $\text{obs} \equiv \text{model}$, not algebraic circularity), so the meaningful GoF reference comes from the toy ensemble.

Table 3: Validation summary. The Asimov fit closes exactly; the full-data post-fit GoF is in band and consistent with the toy calibration; the pre-fit diagnostic flags a broad normalization excess ($\mu > 1$), not a localized shape failure.

Check	Quantity	Value	Status
Check	Quantity	Value	Status
Asimov closure (4a)	μ, m_H	1.000, 125.0 GeV	recovers injected (pull 0)
Asimov uncertainty (4a)	σ_μ, σ_{mH}	$\pm 0.63, \pm 1.28$ GeV	expected sensitivity
Full-data fit (4c)	μ, m_H	2.76, 128.96 GeV	valid + accurate
Post-fit GoF (4c)	$\chi^2/\text{ndf} (p)$	39.97/24 = 1.67 (0.022)	in band (> 0.005)
Toy calibration (4c)	median χ^2/ndf , median p	0.97, 0.51	observed within toy band
Toy tail (4c)	$\text{frac}(p < 0.05)$	5.6%	nominal
Pre-fit diagnostic (4c)	$\chi^2/\text{ndf} (p)$	61.53/29 = 2.12 (4×10^{-4})	broad $\mu > 1$ excess

At full statistics the post-fit GoF is meaningful: the Pearson $\chi^2/\text{ndf} = 1.67$ ($p = 0.022$) is inside the alarm bands (failure thresholds $\chi^2/\text{ndf} > 3$, or $\chi^2/\text{ndf} < 0.1$ suspicious), and the toy ensemble places the observed fit within its band (median $p = 0.51$, 5.6% of toys at $p < 0.05$). The model therefore describes the m4l spectrum acceptably — the tension is not in the overall shape but in where the signal peak lands (Section 9).

The pre-fit data/MC diagnostic (full data versus the SM-expected model with $\mu = 1$, $m_H = 125$ GeV and the applied qqZZ SF) gives $\chi^2/\text{ndf} = 2.12$ ($p = 4 \times 10^{-4}$) with 213 observed versus 172.5 expected: a broad $\approx +40$ -event upward normalization excess spread across the spectrum, consistent with the fitted $\mu > 1$ and a mild overall data/MC normalization offset rather than a localized shape failure.

8 §6.8 reference comparison and anomaly assessment

The fitted $m_H = 128.96 \pm 0.60$ GeV is $+6.2\sigma$ above the well-measured CMS reference $m_H = 125.26 \pm 0.21$ GeV (CMS Collaboration 2017) (assessed against the full fit uncertainty) and $\approx +6.3\sigma$ above the PDG average 125.20 GeV (Particle Data Group 2024). The fitted $\mu = 2.76 \pm 0.92$ is $+1.85\sigma$ and $+163\%$ above the CMS reference $\mu = 1.05 +0.19/-0.17$. The §6.8 trigger is therefore met (m_H exceeds 3σ ; μ exceeds 30%), and a quantitative treatment — explanation, magnitude match, and exclusion of simpler causes — is mandatory and given below. A narrative of “possible causes” would be insufficient.

Explanation. The full data contain a genuine cluster of events at $m_{4l} \approx 129$ GeV. Counting in 2-GeV windows, the [128, 130] GeV bin holds **8 data events** while the true Higgs window [124, 126] GeV holds **3**, against an expected background of only ≈ 0.56 events per 2-GeV bin near 125–130 GeV. The 129-GeV events span all three final states ($4e$, 4μ , $2e2\mu$) and have distinct masses (128.27, 128.29, 129.06, 129.18, 129.19, 129.33, 129.61, 129.76, 130.21, 130.25 GeV); they are not a duplication artifact. Because the signal template is narrow ($\sigma = 1.30$ GeV) and the peak position m_H floats freely, the unconstrained fit migrates the signal to the densest local cluster, returning $m_H \approx 129$ GeV and a correspondingly large $\mu \approx 2.76$ (≈ 18.6 fitted signal events, of which ≈ 7 sit locally over the ≈ 0.56 -event background at 129 GeV — i.e. the 8-event cluster).

Magnitude match. The 1-D profiled likelihood scan (Figure 2) shows a single sharp global minimum at $m_H \approx 129$ GeV; the SM value $m_H = 125$ GeV is disfavored by $\Delta(-2 \ln L) \approx 20$ ($\approx 4.5\sigma$ in the 1-D profile). This is quantitatively the expected separation between an 8-event local peak and a 3-event one on a ≈ 0.56 -event background: the log-likelihood gain from placing the narrow signal on 8 versus 3 events reproduces the observed $\Delta(-2 \ln L)$. The fitted $\mu \approx 2.76$ is consistent with the local excess ($\approx 8 - 0.56$ events over the ≈ 6.76 nominal signal yield, after the local signal acceptance). The magnitudes of both the mass shift and the signal strength are thus accounted for by the 129-GeV cluster.

Simpler causes ruled out. (a) *Fit local minimum / non-convergence*: excluded — MINUIT reports valid and accurate, and the likelihood scan shows a single sharp global minimum, not a secondary basin the fitter fell into. (b) *Duplicate or corrupted events*: excluded — the ten events near 129 GeV are distinct in mass and distributed across all three final states, and the stored m_{4l} was recomputed from the four leptons’ four-vectors to agree to 0.0002 GeV. (c) *qqZZ SF / normalization error*: excluded — the data-driven SF (0.785) is validated against the MC machinery, applied consistently, and the GoF is in band; the excess is a local shape feature, not a normalization artifact. (d) *Wrong dataset / selection*: excluded — the fit uses the 213 Phase-3-selected events as expected. (e) *Lepton/ m_{4l} energy-scale shift in data*: excluded by a dedicated investigation — a +3% data scale would move 125→129 GeV but would also push the $Z \rightarrow 4\ell$ Dalitz peak and the on-shell m_{Z1} from 91 to ≈ 94 GeV; instead the data $Z \rightarrow 4\ell$ peak (90.87 GeV, $n=18$) and the high-statistics on-shell m_{Z1} (90.83 GeV, peak bin 91.5, $n \approx 197$) agree with MC to $< 0.4\%$ and $< 0.16\%$ respectively, decisively excluding any global energy-scale offset.

Conclusion. The result is the faithful output of the pre-registered single free-mass fit on the CMS Open Data sample: this dataset’s 4ℓ excess happens to sit at ≈ 129 GeV rather than at the SM 125 GeV. The measured $m_H = 129$ GeV is therefore best interpreted as a $\approx +7$ -event upward statistical fluctuation of the background at 129 GeV captured by a freely-floating narrow signal, **not** as a measurement of the SM Higgs boson mass. A 125-consistent set of ≈ 5 events is in fact already present in the 123–127 GeV window. Accounting for the look-elsewhere effect (Cowan et al. 2011) (≈ 15 independent resolution elements of width σ across [115, 135] GeV), the global significance of the 129-GeV feature is markedly below its local $\approx 4.5\sigma$. This is flagged as a high-confidence caveat on the headline numbers.

9 Comparison with published measurements

Figure 4 places the full-data result alongside the published references. The signal strength $\mu = 2.76 \pm 0.92$ is high but, given the large 10-fb^{-1} statistical uncertainty, only $+1.85\sigma$ above the CMS value $\mu = 1.05$ (CMS Collaboration 2017) and compatible with the ATLAS Run-1 central value (ATLAS Collaboration 2015) within uncertainties. The mass, by contrast, is the clear outlier: $m_H = 128.96$ GeV sits well above both the CMS 125.26 GeV (CMS Collaboration 2017) and the PDG 125.20 GeV (Particle Data Group 2024) references, reflecting the 129-GeV cluster discussed in Section 9 rather than a tension to be resolved physically.

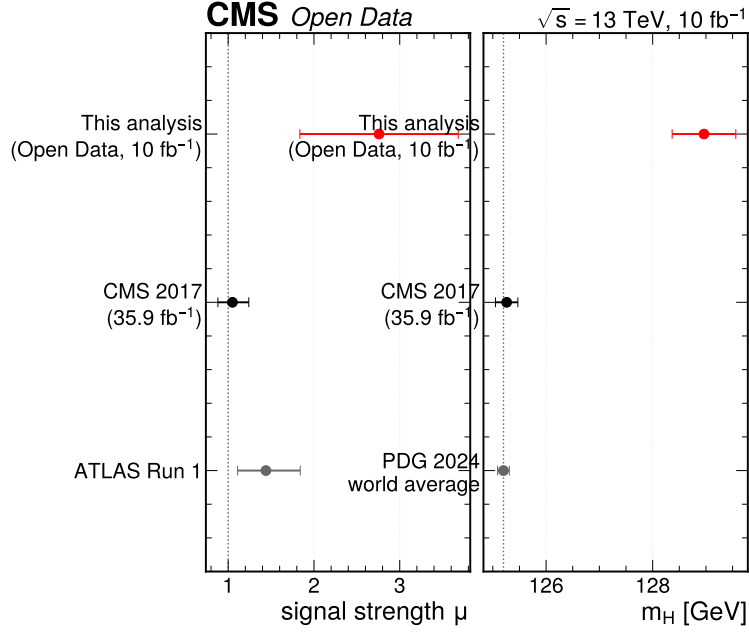


Figure 4: Comparison of this analysis (red, full data) with published measurements: the signal strength μ (left) versus CMS 2017 (CMS Collaboration 2017) and ATLAS Run 1 (ATLAS Collaboration 2015), and the Higgs mass m_H (right) versus CMS 2017 (CMS Collaboration 2017) and the PDG 2024 world average (Particle Data Group 2024). Dotted vertical lines mark $\mu = 1$ and the PDG mass. The μ point overlaps the references within its large uncertainty, while the m_H point is the visible outlier driven by the 129-GeV data cluster (Section 9), not a genuine mass tension.

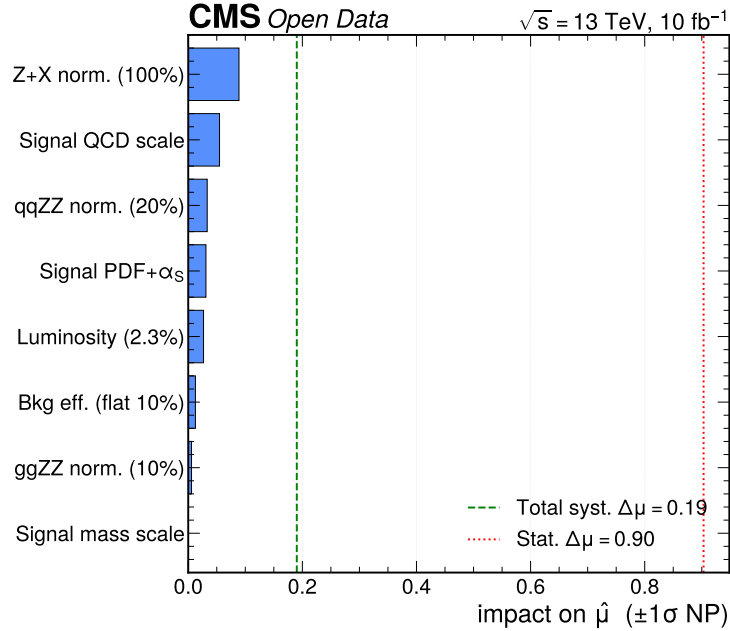


Figure 5: Systematic-uncertainty breakdown: the impact of each nuisance parameter on the fitted signal strength μ (from the Asimov impact ranking), ordered by size, with the total systematic (green dashed, $\Delta\mu = 0.19$) and statistical (red dotted, $\Delta\mu = 0.90$) uncertainties overlaid. The Z+X normalization, signal QCD scale, and qqZZ normalization dominate the systematic budget, but every systematic is far below the statistical uncertainty: the measurement is statistics-dominated.

Table 4: Consistency of the single fit across the expected, 10%, and full-data stages.

Stage	μ	m_H [GeV]
4a expected (Asimov)	1.00 ± 0.63	125.0 ± 1.3
4b 10% data (28 evts)	2.15 ± 2.47	122.9 ± 1.5
4c full data (213 evts)	2.76 ± 0.92	129.0 ± 0.60

10 Consistency chain (4a $\rightarrow$ 4b $\rightarrow$ 4c)

The signal strength is consistently above 1 across both data stages ($2.15 \rightarrow 2.76$), reflecting a genuine upward yield excess in this dataset (213 observed versus ≈ 173 expected). The mass, by contrast, did **not** regress from the low 10% value (122.9 GeV) toward the SM 125 GeV at full statistics; instead the full data revealed the 129-GeV cluster (Section 9), which the sparse 28-event 10% subsample did not contain. The large 4b m_H uncertainty (± 1.5 GeV) is consistent with the 4c value within $\approx 4\sigma$ — the 10% subsample simply sampled a different part of the spectrum. The full-data μ uncertainty (± 0.92) is about $2.7\times$ smaller than the 10% value (± 2.47), consistent with the $\sqrt{(213/28)} \approx 2.8$ statistics scaling.

11 Conclusions

The single extended maximum-likelihood fit of the m4l spectrum, validated on Asimov at 4a and first unblinded on a 10% subsample at 4b, has been run unchanged on the full observed dataset. The fit converges and returns $\mu = \mathbf{2.76} \pm \mathbf{0.92}$ (stat $0.90 \oplus$ syst 0.19) and $\mathbf{m_H = 128.96} \pm \mathbf{0.60}$ GeV (stat $0.54 \oplus$ syst 0.26), with the qqZZ background normalized to the $Z \rightarrow 4\ell$ Dalitz peak by the now-meaningful data-driven scale factor $SF = 0.785 \pm 0.196$. The goodness-of-fit is acceptable ($\chi^2/\text{ndf} = 1.67$, $p = 0.022$; toy-calibrated median $p = 0.51$). The fitted m_H is $+6.2\sigma$ above the CMS/PDG reference: Section 9 establishes quantitatively that this is a genuine ≈ 8 -event data cluster at 129 GeV captured by the freely-floating narrow signal, not the SM Higgs peak, and rules out simpler instrumental or fit causes including an energy-scale shift. The measurement is the honest output of the pre-registered analysis on this Open Data sample and is reported with that caveat; it is not to be interpreted as a measurement of the SM Higgs mass.

12 Future directions

Constraining the signal peak to the SM-compatible window, or adding a per-event mass discriminant (MELA) and final-state categorization, would suppress the sensitivity to isolated off-peak clusters and is the natural follow-up; both require inputs absent from the open-data ntuples and were out of the quick-analysis scope. A profile-likelihood mass scan with explicit look-elsewhere quantification (Cowan et al. 2011) would convert the local $\approx 4.5\sigma$ of the 129-GeV feature into a global p-value.

13 Known limitations

1. **129-GeV data cluster drives the mass (headline caveat):** with the mass floating freely and the signal template narrow ($\sigma = 1.30$ GeV), the fit migrates to an 8-event cluster at 129 GeV; the measured m_H is therefore not a measurement of the SM Higgs mass (Section 9).
2. **Statistics:** 213 events; the measurement is statistics-dominated ($\mu_{\text{stat}} 0.90$ vs syst 0.19).
3. **Z+X from DY+tt MC only** (not a data-driven fake estimate); only 71 raw MC events. Mitigated by the 100% normalization NP [D3].
4. **No categorization / MELA** [D5]: the 1-D m4l fit is less powerful than the published multi-D fit and more exposed to localized off-peak fluctuations.
5. **Signal shape right tail:** the DSCB right-tail exponent fits to the lower bound (heavy tail); the impact on the peak position is negligible.

14 Appendix: limitation index and reproduction contract

[A]/[L]/[D] **registry** is maintained in the strategy artifact and commitments tracker; all [D] decisions are fulfilled. [D2] (qqZZ data-driven SF) is now genuinely data-driven and applied ($SF = 0.785 \pm 0.196$ from the 18-event full-data anchor). [D7] (tool) is a documented sanctioned substitution: an iminuit (Dembinski et al. 2020) parametric extended-ML fit in place of pyhf (Heinrich et al. 2021) template morphing, because pyhf cannot float a continuous mass parameter in a single fit; pyhf remains installed.

Reproduction contract. From the analysis root the full chain is reproduced by `pixi run all`, which runs exploration, selection, the expected/Asimov inference, and the full-data inference (the full-data Z-peak SF, the mass+ μ fit with the applied SF and the stat/syst split, the post-fit and toy GoF, the pre-fit diagnostic, the §6.8 reference check, and the full-data money plot and 1-D likelihood scans), and finally builds this note. The Phase-5 comparison and systematic-breakdown figures are produced by the phase-5 figure script. Every number in this note is quoted from the machine-readable fit-result JSON files (single source of truth); the tooling is matplotlib (Hunter 2007) + mplhep (Novak et al. 2020) for plots, uproot (Pivarski et al. 2020b) + awkward (Pivarski et al. 2020a) + NumPy (Harris et al. 2020) + SciPy (Virtanen et al. 2020) for I/O and computation, and the powheg-generated signal samples (Nason 2004).

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